AZURE CLOUD COMPUTING

Virtual Machine and Linux Administration Assignment

Student	Anhar Taha Al-Rujaee
Platform	Microsoft Azure
Operating System	Ubuntu Server 24.04 LTS
Virtual Machine	AnharAlRujaee-VM
Region	South Africa North
Evidence date	7 November 2026

Practical implementation report

Executive Summary

This report presents the setup and practical administration of an Ubuntu virtual machine in Microsoft Azure. The work covers Apache web hosting, MySQL database administration, Samba file-sharing configuration, and Azure File Share integration.

1. Introduction

This report documents a practical assignment in Microsoft Azure and Linux administration. The work involved deploying an Ubuntu virtual machine, publishing a website with Apache, creating a MySQL database, configuring Samba file sharing, and mounting an Azure File Share for file operations.

2. Azure Virtual Machine Setup

An Ubuntu Server 24.04 virtual machine named AnharAlRujaee-VM was created in Microsoft Azure. The VM overview screenshot records the region as South Africa North and shows the VM in the Running state. A remote shell session was established from a Mac terminal using SSH. Public IP and subscription identifiers are intentionally omitted from this public report.

3. Task 1 – Apache Web Server and Website

Apache was installed using `sudo apt update` and `sudo apt install apache2 -y`. The service was enabled and started with `sudo systemctl enable --now apache2`, and its status was checked using `sudo systemctl status apache2`. A custom HTML welcome page was placed in `/var/www/html/index.html`. Initially, the website timed out because inbound HTTP traffic was not allowed by the Azure Network Security Group. An inbound rule allowing TCP port 80 was added (priority 310), after which the website became accessible through the VM public IP.

4. Task 2 – MySQL Database

MySQL Community Server 8.0.46 was installed and enabled. The service status showed `active (running)`. Using `sudo mysql`, the database `anhar_database` was created, followed by a `students` table with `id`, `name`, and `course` columns. A sample record (Anhar, Linux Administration) was inserted and retrieved using `SELECT * FROM students;`. The query returned one row successfully.

5. Task 3 – Samba File Sharing

Samba was installed and the service `smbd` was confirmed active and running. A shared directory was created at `/srv/samba/shared`, a `sambashare` group was created, and the Ubuntu user was added to that group. Directory ownership and permissions were configured, and a Samba password was set. The `[AnharShare]` share was configured in `/etc/samba/smb.conf` with browseable access, write access, group-based valid users, and file/directory creation masks. The configuration was validated with `testparm` according to the completed setup workflow. A separate connection test from another client was not captured and is not claimed in this report.

6. Task 4 – Azure File Share

A storage account named `nexuscrm` and a file share named `anhar-files` were created. The share was mounted on the Ubuntu VM at `/media/anhar-files`, and the mount was verified with `df -h`. A test file named `test.txt` was created using `tee`, its contents were displayed using `cat`, and it was deleted using `rm`. A final directory listing showed the share empty again. These operations demonstrate successful create, read, and delete access to the mounted Azure File Share.

7. Problems Encountered and Solutions

The published website initially returned a connection timeout. The cause was an Azure Network Security Group rule that did not allow inbound HTTP. An Allow-HTTP rule for TCP port 80 was added and the website then worked. Samba required installation, a dedicated shared directory, group permissions, a Samba user password, and a share definition in the configuration file. The Azure File Share mount was checked with `df -h` before performing file operations.

8. Additional Technical Training and Projects

The student reports technical learning in Python programming, PostgreSQL databases, and UI design through IT Bootcamp. Additional learning/development activities are associated with VIT Academy and We RHere IT Academy. Separately, the student designed and developed NexusCRM, a CRM desktop application using Python, PyQt6, and PostgreSQL. The NexusCRM screenshots are included in a separate supplementary folder and are not presented as evidence for the Azure tasks.

9. Conclusion

The four main practical tasks were completed: Apache web hosting, MySQL database creation and query execution, Samba share configuration, and Azure File Share mounting with create/read/delete operations. The assignment provided practical experience with Azure VM deployment, SSH, Linux services, network security rules, relational databases, and cloud file storage.

10. Submission Links

GitHub repository:
https://github.com/AnharAlRujaee/Azure-Cloud-Computing-Virtual-Machine-VM-and-Linux-Administration
LinkedIn profile: https://www.linkedin.com/in/anhar-taha-al-rujaee-645668226/
LinkedIn post URL: To be added after publication.

Appendix A – Practical Screenshots

Azure VM overview

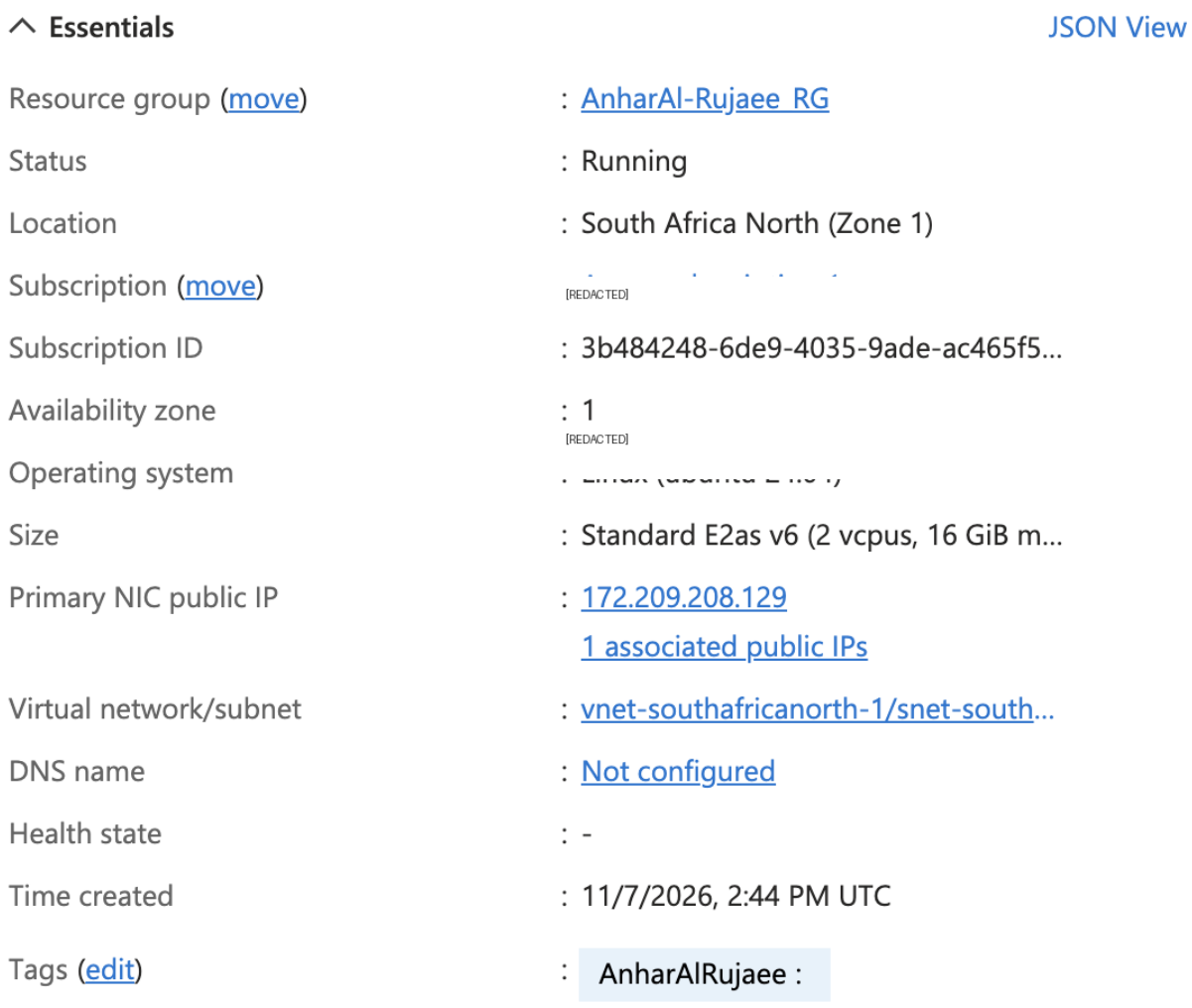


Figure: Azure VM overview

SSH session and Ubuntu system information

```
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 7.0.0-1017-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Nov  7 15:24:57 UTC 2026

  System load:  0.16              Temperature:           49.9 C
  Usage of /:   6.6% of 28.02GB   Processes:             139
  Memory usage: 2%                Users logged in:       0
  Swap usage:   0%                IPv4 address for eth0: 172.16.0.4
```

Figure: SSH session and Ubuntu system information

Apache service status

```
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
     Active: active (running) since Sat 2026-11-07 15:52:54 UTC; 43s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2744 (apache2)
      Tasks: 55 (limit: 17441)
     Memory: 5.3M (peak: 5.6M)
        CPU: 28ms
     CGroup: /system.slice/apache2.service
             ├─2744 /usr/sbin/apache2 -k start
             ├─2746 /usr/sbin/apache2 -k start
             └─2747 /usr/sbin/apache2 -k start
```

Figure: Apache service status

Published Apache website



Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in /usr/share/doc/apache2/README.Debian.gz**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|       `--  ports.conf
|-- mods-enabled
|       |-- *.load
|       `-- *.conf
|-- conf-enabled
|       `-- *.conf
|-- sites-enabled
|       `-- *.conf
```

- `apache2.conf` is the main configuration file. It puts the pieces together by including all remaining configuration files when starting up the web server.
- `ports.conf` is always included from the main configuration file. It is used to determine the listening ports for incoming connections, and this file can be customized anytime.
- Configuration files in the `mods-enabled/`, `conf-enabled/` and `sites-enabled/` directories contain particular configuration snippets which manage modules, global configuration fragments, or virtual host configurations, respectively.
- They are activated by symlinking available configuration files from their respective *-available/ counterparts. These should be managed by using our helpers `a2enmod`, `a2dismod`, `a2ensite`, `a2dissite`, and `a2enconf`, `a2disconf` . See their respective man pages for detailed information.
- The binary is called apache2 and is managed using systemd, so to start/stop the service use `systemctl start apache2` and `systemctl stop apache2`, and use `systemctl status apache2` and `journalctl -u apache2` to check status. `system` and `apache2ctl` can also be used for service management if desired. **Calling `/usr/bin/apache2` directly will not work** with the default configuration.

Document Roots

By default, Ubuntu does not allow access through the web browser to *any* file outside of those located in `/var/www`, **public_html** directories (when enabled) and `/usr/share` (for web applications). If your site is using a web document root located elsewhere (such as in `/srv`) you may need to whitelist your document root directory in `/etc/apache2/apache2.conf`.

The default Ubuntu document root is `/var/www/html`. You can make your own virtual hosts under `/var/www`.

Reporting Problems

Please use the `ubuntu-bug` tool to report bugs in the Apache2 package with Ubuntu. However, check **existing bug reports** before reporting a new bug.

Please report bugs specific to modules (such as PHP and others) to their respective packages, not to the web server itself.

Welcome to Anhar's Azure VM!

My Apache web server is running successfully.
Hosted on Ubuntu 24.04 LTS.

Figure: Published Apache website

Azure inbound HTTP security rule

Network security group AnharAlRujaee-vm-nsg (attached to networkInterface: anharalrujaee-vm541)
Impacts 0 subnets, 1 network interfaces

Create port rule

Search rules | Source == all | Destination == all | Protocol == all | Action == all | Port == all

Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (5)						
300	SSH	22	TCP	Any	Any	Allow
310	Allow-HTTP	80	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Figure: Azure inbound HTTP security rule

MySQL service status

```
[AnharAlRujaee@AnharAlRujaee-vm:~$ sudo systemctl status mysql                  ]
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: en>
     Active: active (running) since Sat 2026-11-07 16:05:25 UTC; 29s ago
   Main PID: 4212 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 17441)
     Memory: 363.7M (peak: 378.0M)
        CPU: 700ms
     CGroup: /system.slice/mysql.service
             └─4212 /usr/sbin/mysqld

Nov 07 16:05:24 AnharAlRujaee-vm systemd[1]: Starting mysql.service - MySQL Com>
Nov 07 16:05:25 AnharAlRujaee-vm systemd[1]: Started mysql.service - MySQL Comm>
[AnharAlRujaee@AnharAlRujaee-vm:~$ sudo mysql                                   ]
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46-0ubuntu0.24.04.4 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.
```

Figure: MySQL service status

MySQL database, table and query result

```
mysql> CREATE DATABASE anhar_database;
Query OK, 1 row affected (0.00 sec)

mysql> USE anhar_database;
Database changed
mysql> CREATE TABLE students (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     course VARCHAR(100)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO students (name, course)
    -> VALUES ('Anhar', 'Linux Administration');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM students;
+----+-------+----------------------+
| id | name  | course               |
+----+-------+----------------------+
|  1 | Anhar | Linux Administration |
+----+-------+----------------------+
1 row in set (0.00 sec)

mysql> EXIT;
Bye
```

Figure: MySQL database, table and query result

Samba service status

```
AnharAlRujaee@AnharAlRujaee-vm:~$ sudo systemctl status smbd
● smbd.service - Samba SMB Daemon
     Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: ena>
     Active: active (running) since Sat 2026-11-07 16:08:42 UTC; 11s ago
       Docs: man:smbd(8)
             man:samba(7)
             man:smb.conf(5)
   Main PID: 5484 (smbd)
     Status: "smbd: ready to serve connections..."
      Tasks: 3 (limit: 17441)
     Memory: 10.4M (peak: 10.6M)
        CPU: 47ms
     CGroup: /system.slice/smbd.service
             ├─5484 /usr/sbin/smbd --foreground --no-process-group
             ├─5487 "smbd: notifyd" .
             └─5488 "smbd: cleanupd "

Nov 07 16:08:42 AnharAlRujaee-vm systemd[1]: Starting smbd.service - Samba SMB >
Nov 07 16:08:42 AnharAlRujaee-vm (smbd)[5484]: smbd.service: Referenced but uns>
Nov 07 16:08:42 AnharAlRujaee-vm systemd[1]: Started smbd.service - Samba SMB D>
```

Figure: Samba service status

Samba installation output

```
AnharAlRujaee@AnharAlRujaee-vm:~$ sudo apt update
[sudo apt install samba -y                                                    ]
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  attr libavahi-client3 libavahi-common-data libavahi-common3
  libboost-iostreams1.83.0 libboost-locale1.83.0 libboost-thread1.83.0
  libcephfs2 libcups2t64 libldb2 librados2 librdmacm1t64 libtdb1 libtevent0t64
  liburing2 python3-dnspython python3-gpg python3-ldb python3-markdown
  python3-samba python3-talloc python3-tdb samba-ad-provision samba-common
  samba-common-bin samba-dsdb-modules samba-libs samba-vfs-modules tdb-tools
Suggested packages:
  cups-common python3-trio python3-aioquic python3-h2 python3-httpx
  python3-httpcore python-markdown-doc bind9 bind9utils ctdb ldb-tools winbind
  heimdal-clients
The following NEW packages will be installed:
  attr libavahi-client3 libavahi-common-data libavahi-common3
  libboost-iostreams1.83.0 libboost-locale1.83.0 libboost-thread1.83.0
  libcephfs2 libcups2t64 libldb2 librados2 librdmacm1t64 libtdb1 libtevent0t64
  liburing2 python3-dnspython python3-gpg python3-ldb python3-markdown
  python3-samba python3-talloc python3-tdb samba samba-ad-provision
  samba-common samba-common-bin samba-dsdb-modules samba-libs
  samba-vfs-modules tdb-tools
0 upgraded, 30 newly installed, 0 to remove and 14 not upgraded.
Need to get 19.6 MB of archives.
After this operation, 105 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython
all 2.6.1-1ubuntu1 [163 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libtdb1 amd64 1.4.
10-1build1 [46.8 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libtevent0t64 amd6
4 0.16.1-2build1 [42.6 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libldb2 am
d64 2:2.8.0+samba4.19.5+dfsg-4ubuntu9.8 [188 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-ld
b amd64 2:2.8.0+samba4.19.5+dfsg-4ubuntu9.8 [41.8 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 python3-tdb amd64
```

Figure: Samba installation output

Azure File Share mount, create, read and delete

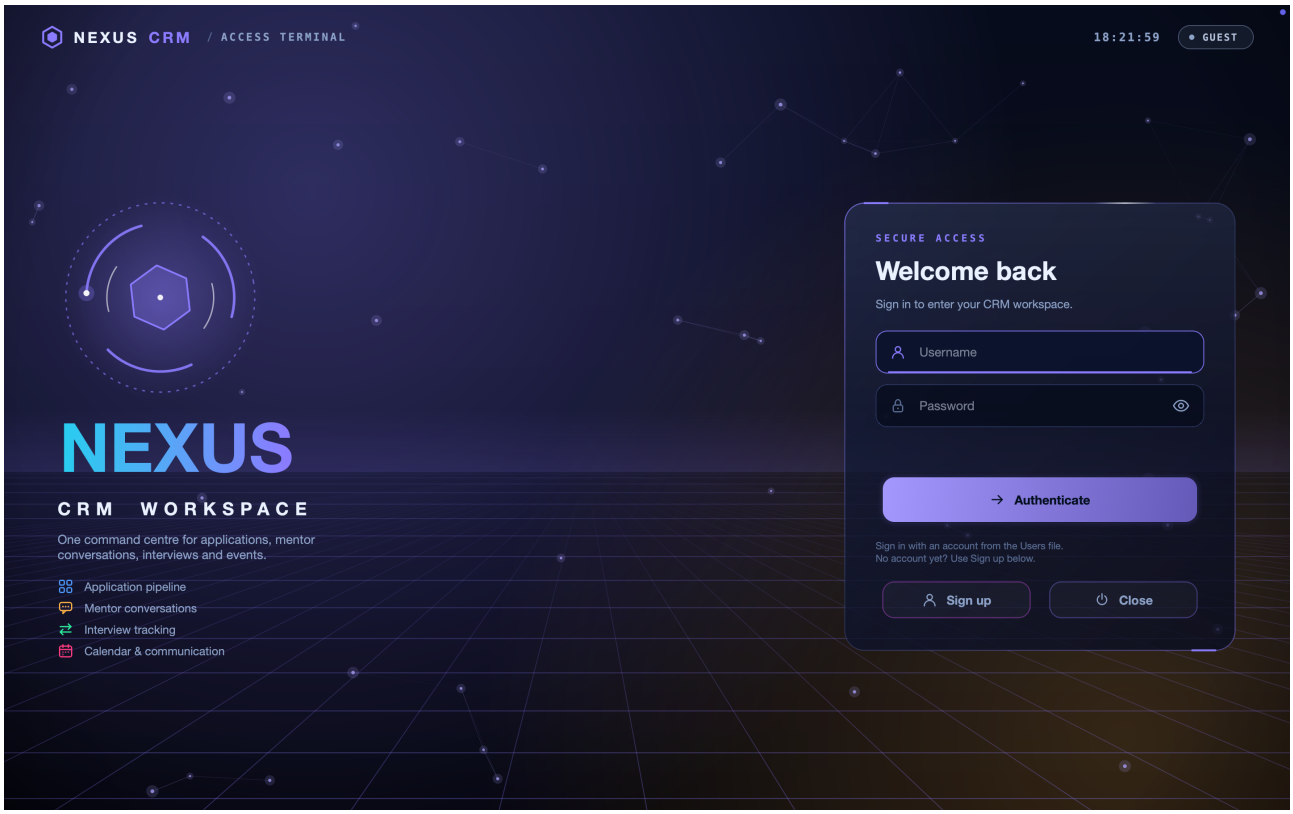
```
AnharAlRujaee@AnharAlRujaee-vm:~$ df -h /media/anhar-files
Filesystem                                      Size  Used Avail Use% Mounted on
//nexuscrm.file.core.windows.net/anhar-files  100T     0  100T   0% /media/anhar
-files
AnharAlRujaee@AnharAlRujaee-vm:~$ ls -la /media/anhar-files
total 4
drwxr-xr-x 2 root root    0 Nov  7 15:46 .
drwxr-xr-x 3 root root 4096 Nov  7 15:46 ..
AnharAlRujaee@AnharAlRujaee-vm:~$ echo "Hello! This file was created from my Azu
re Ubuntu VM." | sudo tee /media/anhar-files/test.txt
Hello! This file was created from my Azure Ubuntu VM.
AnharAlRujaee@AnharAlRujaee-vm:~$ cat /media/anhar-files/test.txt
Hello! This file was created from my Azure Ubuntu VM.
AnharAlRujaee@AnharAlRujaee-vm:~$ sudo rm /media/anhar-files/test.txt
AnharAlRujaee@AnharAlRujaee-vm:~$ ls -la /media/anhar-files
total 4
drwxr-xr-x 2 root root    0 Nov  7 15:46 .
drwxr-xr-x 3 root root 4096 Nov  7 15:46 ..
```

Figure: Azure File Share mount, create, read and delete

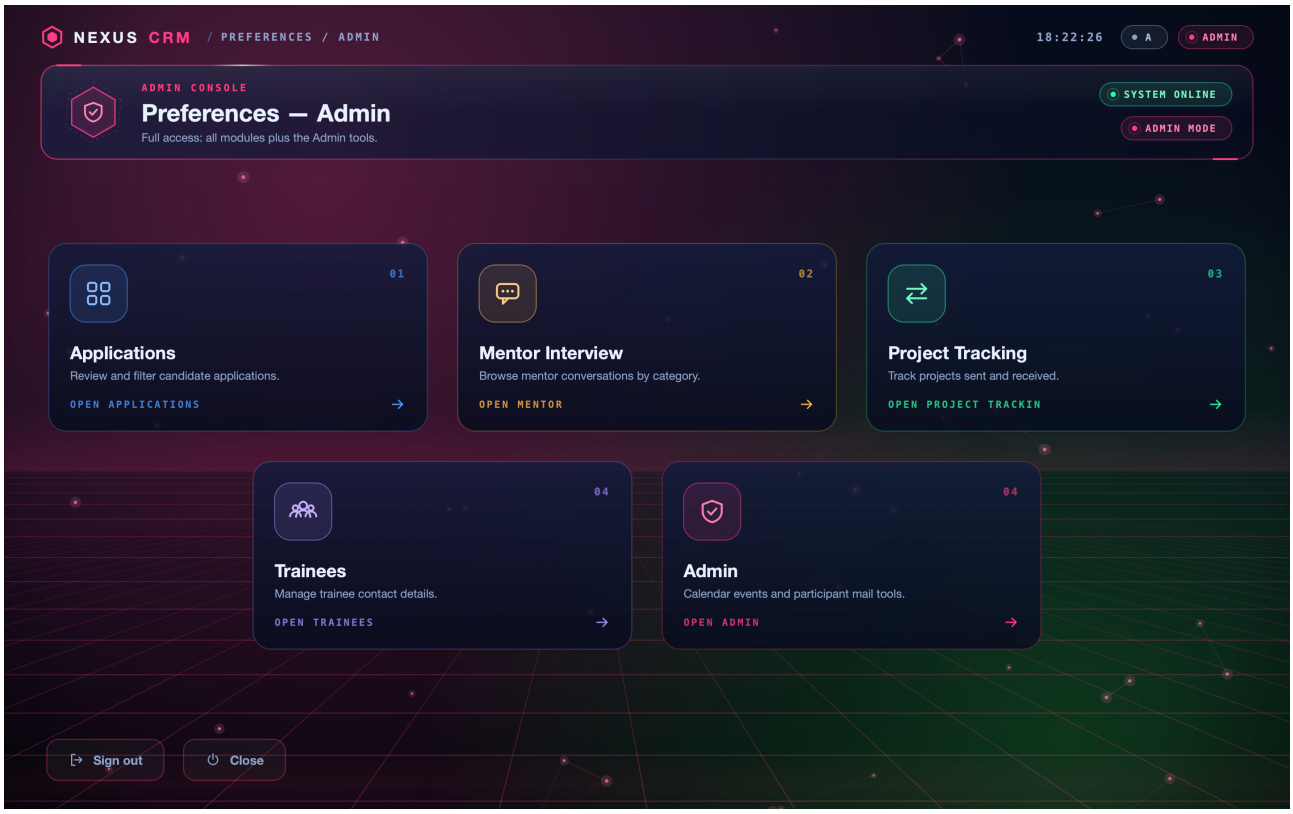
Appendix B – Supplementary NexusCRM Project

These images are a separate visual showcase of NexusCRM, a CRM application developed with Python, PyQt6, and PostgreSQL. They are supplementary portfolio material and are not part of the Azure assignment evidence.

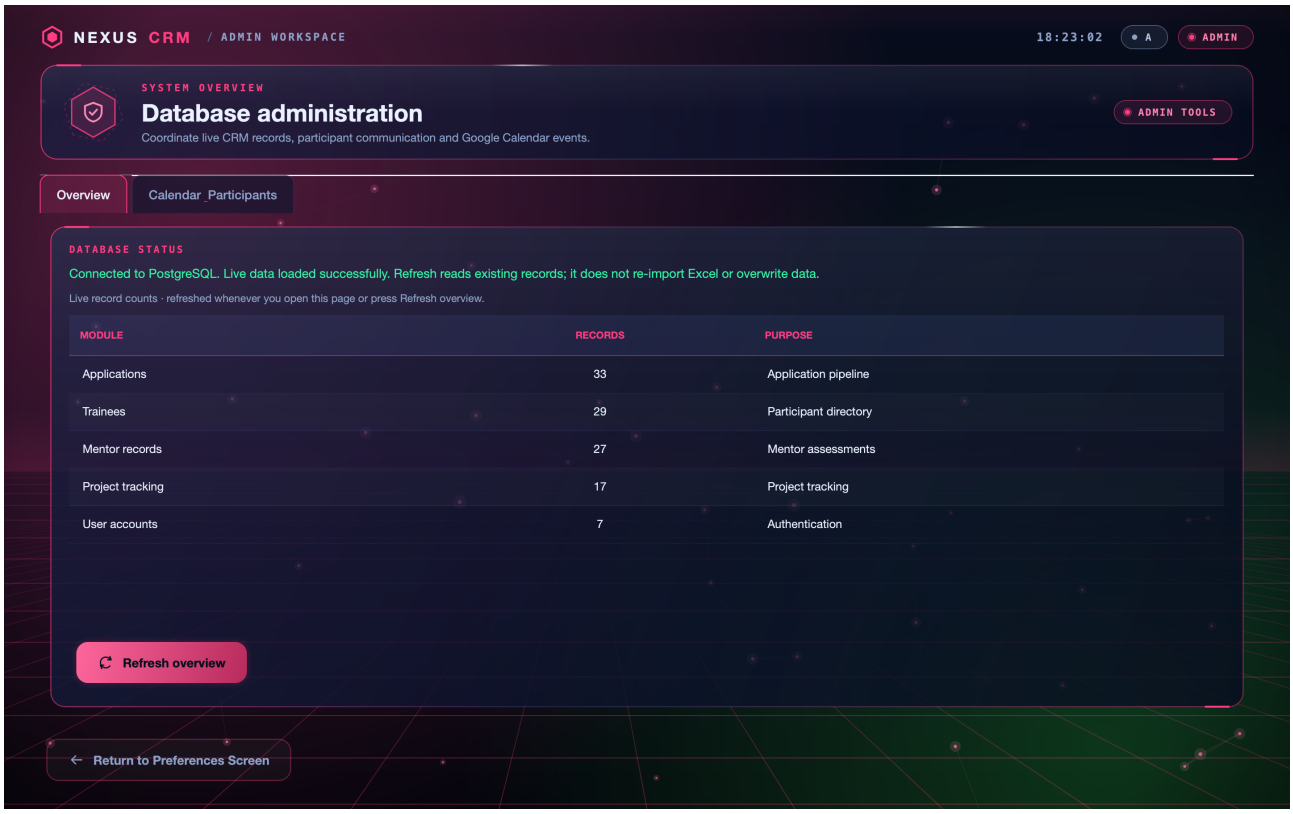
01 NexusCRM Login



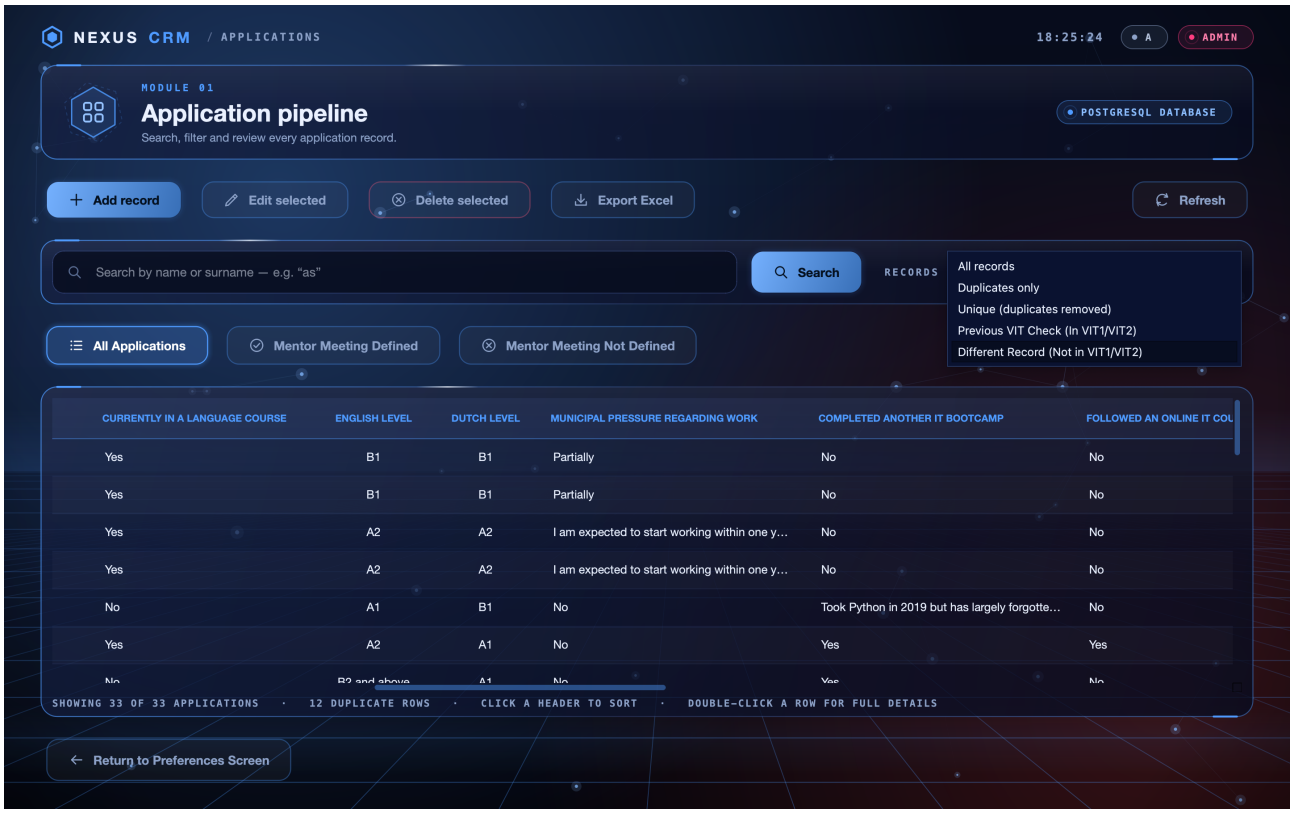
02 NexusCRM Admin Dashboard



03 NexusCRM Database Overview



04 NexusCRM Application Pipeline



05 NexusCRM Code Development

